SHAFIQ MAHMUD

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PROFESSIONAL SUMMARY

Undergraduate researcher in Urban and Regional Planning at CUET, applying GIS, remote sensing, and machine learning to flood, landslide, drought, and storm hazard susceptibility and risk assessment. Co-author of two preprints and a conference paper, with an ongoing thesis on machine learning-based agricultural drought risk assessment and further work on uncertainty-aware modeling of coastal storm impacts. Seeking graduate study to advance uncertainty-aware, geospatial approaches to disaster-resilient planning.

EDUCATION

Bachelor of Urban and Regional Planning (BURP) 2023–Present
Chittagong University of Engineering and Technology (CUET) | CGPA: 3.35 (up to 6th Semester)
Higher Secondary Certificate (HSC), Dhaka College | GPA: 5.00 2021
Secondary School Certificate (SSC), Basurhat A.H.C Government High School | GPA: 5.00 2019

RESEARCH & PUBLICATIONS

- **Preprint (EarthArXiv, May 8, 2026):** "Enhancing the Generalization of Flood Susceptibility Models: A Leakage-Aware Ensemble Framework for Deltaic Landscapes." **Shafiq Mahmud**, Golam Murad, Md. Aftabur Rahman.
- **Preprint (EarthArXiv, Mar 26, 2026):** "Uncertainty-Aware Bayesian Machine Learning for Landslide Susceptibility Mapping in the Chattogram Metropolitan Hill System, Bangladesh." Golam Murad, Md. Aftabur Rahman, Shotabdy Chowdhury Srabony, **Shafiq Mahmud**.
- **Conference Paper (Presented):** "National-Scale Drought Mapping Using Satellite-Derived Biophysical Indices: Towards a Robust Drought Monitoring System in Bangladesh." International Conference on Civil Engineering Research & Innovations, RUET, Dec 12–14, 2025. Sajib Sarker, **Shafiq Mahmud**, Akifa Monjur, Md. Abdullah Talbi, Joy Sikder.
- **Undergraduate Thesis (Ongoing):** "Machine Learning-Based Agricultural Drought Risk Assessment in Bangladesh." Department of Urban and Regional Planning, CUET.
- **Manuscript (Ongoing):** "Uncertainty-informed bivariate decision zones for storm-impact susceptibility in the cyclone-exposed southwestern coast of Bangladesh." Anindya Baidya, Anik Chowdhury, **Shafiq Mahmud**, Md. Aftabur Rahman*, Tanvir Ahmed. (*Corresponding author)
- **Manuscript (Ongoing):** "Coastal Land Dynamics of Noakhali, Bangladesh (2000–2020): A Multi-Sensor Remote Sensing Analysis and Engineering Assessment for Economic Zone Development."
- **Investigative Report (Unpublished):** "Fragile Slopes, Damaged Roads: An Investigation into Clustered Landslides along the Rajasthali-Saichal Segment of the Border Road, Bangladesh." Golam Murad, Md. Aftabur Rahman, **Shafiq Mahmud**.

ACADEMIC PROJECTS

- **Flood Susceptibility Mapping in Noakhali Using Artificial Neural Networks** – Advanced GIS Lab (URP 366)
- **Spatio-temporal Assessment of Nighttime Light Emissions and Urban Vulnerability: A Case Study of Dhaka for the Years 2015, 2020 and 2025**– Remote Sensing and Photogrammetry Lab (URP 364)
- **Spatial-Seasonal Dynamics of Air Quality and Public Health Vulnerability in Chattogram City Corporation, Bangladesh** – Climate and Disaster Management Sessional (URP 436)

TECHNICAL SKILLS

- **Geospatial & Remote Sensing:** ArcGIS Pro, ArcMap, QGIS, ERDAS Imagine, Google Earth Engine
- **Programming & Modeling:** Python, R, JavaScript; machine learning, susceptibility modeling, uncertainty-aware methods
- **Other:** AutoCAD, SketchUp, Microsoft Office, Adobe Photoshop/Lightroom/Premiere Pro
- **Languages:** Bangla, English

HONORS & AWARDS

- **First Prize** – A Carbon Conscious Urban Design Idea Competition (UNOPS, RAJUK, MoHPW; support from BMZ), concept "The Receding Membrane"
- **Winner** – Inter-University Quiz on Spatial Planning Framework for Bangladesh, Bangladesh Institute of Planners (Nov 2024)
- **First Prize** – World Town Planning Day 2025 Photography Contest on Anticipatory Planning for Disaster Risk Reduction, CUET (Nov 2025)

PROFESSIONAL & FIELD EXPERIENCE

- **Industrial Attachment, Dhaka North City Corporation (DNCC):** Team-based study of ward-wise community facilities, compared against Private Residential Land Development Rules (PRLDR) standards.
- **Field Visit, Cox's Bazar Development Authority and Khurushkul Special Ashrayan Project:** Observed urban management works and a large-scale climate-migrant rehabilitation initiative.

LEADERSHIP & OTHER ACTIVITIES

- President, CUET Photographic Society; Vice-President, Planning Club, CUET (2026–27); Organizing Committee Member, TEDxCUET (2025)
- Photography/videography (incl. drone) for YPSA, RIMES, and Save the Children; creator of a YouTube channel on GIS & Remote Sensing tutorials